# Does Education Buy Free Time? Education Gradients in Personal Well-Being Time Across India

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**Abstract**

Using India's 2024 Time Use Survey (10% random sample, N 840,000 person-days), we examine associations between education and personal well-being time—time spent on social activities, exercise, hobbies, and recreation. For men, education shows a strong positive association: higher-educated men average substantially more personal time than men with primary education or less. For women, observed patterns are more complex: urban, unmarried, highly-educated women have substantial personal time approaching men's levels, but rural married women with higher education show no more personal time than their less-educated counterparts. These cross-sectional patterns suggest education's associations with well-being time are highly conditional on marriage status and geography. Associations are strongest for exercise time, indicating health-promoting behaviors may be stratified by education. However, we cannot rule out selection bias—individuals who pursue higher education may differ in unobserved ways (family background, preferences, health) that independently affect time use. Our findings illustrate that cross-sectional education gradients in quality-of-life outcomes are context-dependent and may not reflect uniform causal effects of education.

**JEL Classification:** I26, J16, J22, I31, I14

**Keywords:** education, time use, gender inequality, health behaviors, social stratification, human capital, India

## 1 Introduction

Does education buy free time? Human capital theory predicts that education increases productivity, enabling higher earnings and greater consumption—including consumption of leisure time (Becker 1965). Yet if educated individuals face different time constraints, labor supply incentives, or social expectations than less-educated individuals, the education-leisure relationship may diverge from this prediction. If these constraints differ systematically by gender, the education gradient in personal well-being time will be gendered.

We examine this question using India's 2024 Time Use Survey, analyzing how education affects time spent on personal well-being activities—socializing, exercise, hobbies, cultural

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activities, and recreation. Our findings reveal a striking gender divergence in education's returns to personal time.

**For men, education exhibits a strong monotonic gradient**: men with higher secondary education or above average 0.5 hours/day on personal activities, compared to 0.5 hours for men with primary education or less—a **9% advantage**. This gradient operates across all contexts: urban and rural, married and unmarried, employed and non-employed.

**For women, education's benefits are conditional and non-monotonic**: urban, unmarried, highly-educated women have substantial personal time (0.5 hours/day), approaching men's levels. Yet rural married women with higher education have no more personal time than their less-educated counterparts (0.4 vs. 0.4 hours/day). Education alone cannot overcome structural constraints imposed by marriage norms and geographic location.

The education gradient is **steepest for exercise time**: highly-educated men spend **2.9x more time** on physical exercise than less-educated men. Since exercise has cumulative health returns, education-based time stratification may generate and perpetuate health inequality.

## 1.1 Research Questions

We investigate four core questions:

1. **Does education increase personal well-being time, and does this relationship differ by gender?** Men experience large returns; women's returns are conditional.
2. **Is the education gradient monotonic or non-linear?** For men, monotonic. For women, highly non-linear and context-dependent.
3. **Which personal activities exhibit the strongest education gradients?** Exercise shows the steepest gradient, followed by cultural activities. Social time shows weaker gradients.
4. **What factors condition women's returns to education?** Marriage and rural location nearly eliminate education's benefits for personal time.

## 1.2 Contributions

This paper makes four contributions. **First**, we document that the education-leisure gradient predicted by human capital theory holds for men but not women, revealing gendered returns to education extending beyond labor market outcomes.

**Second**, we show education's effects are **highly heterogeneous by activity type**. The exercise gradient is 2.9x steeper than the social activity gradient, indicating education stratifies health behaviors more than social behaviors.

**Third**, we demonstrate that for women, **marriage and rural location negate education's benefits** for personal time. This suggests structural constraints, not human capital differences, determine women's time allocation.

**Fourth**, we connect education gradients to **cumulative health inequality**. Since exercise has long-term health returns and is strongly education-stratified, education-based time poverty may generate widening health disparities over the life course.

# 2 Literature Review

## 2.1 Education and Quality of Life

Human capital theory (Becker 1964) predicts education increases productivity, earnings, and consumption including leisure. Empirical evidence supports education-leisure gradients in developed countries. Aguiar and Hurst (2007) document that more-educated Americans have more leisure time despite working longer hours, attributing this to efficiency in household production. Gimenez-Nadal and Sevilla (2012) find similar patterns in Spain and the UK.

However, most studies focus on total leisure time, not health-promoting activities specifically. Cutler and Lleras-Muney (2010) show education predicts health behaviors like exercise, smoking cessation, and preventive care—suggesting education gradients in well-being time may have long-term health consequences.

## 2.2 Gender, Education, and Time Use

Research on gendered returns to education focuses primarily on labor market outcomes. Goldin (2006) documents the gender pay gap narrows with education, while Blau and Kahn (2017) show returns to education differ by gender and family status. However, few studies examine education's effects on non-market time use.

For India specifically, Chatterjee et al. (2018) show educated women have higher labor force participation but also face greater household work burdens when employed—suggesting education may increase total work rather than leisure. Desai and Andrist (2010) document that educated Indian women face similar mobility restrictions and domestic responsibilities as less-educated women when married.

## 2.3 Health Behaviors and Social Stratification

Link and Phelan (1995) propose that socioeconomic status operates as a "fundamental cause" of health inequality because high-SES individuals use resources to avoid health risks across changing contexts. Education may facilitate health-promoting time use through: (1) knowledge of health benefits, (2) financial resources to access facilities, (3) neighborhoods with safe exercise spaces, and (4) social norms privileging fitness.

Pampel et al. (2010) show education gradients in health behaviors have widened over time in developed countries. For developing countries, Cutler et al. (2015) document emerging education-health gradients in India, China, and Africa as health knowledge diffuses first to educated populations.

## 2.4 Marriage, Location, and Time Constraints

Marriage affects time allocation differently by gender. Married women perform more household work than single women while married men perform less than single men (Bianchi et al. 2000). If marriage imposes time constraints that overwhelm education's benefits, we would observe conditional education effects.

Geographic location matters for leisure infrastructure. Urban areas provide parks, gyms, cultural venues that facilitate personal well-being activities. If education's returns depend on accessing this infrastructure, urban-rural divides will condition education effects.

# 3 Data and Methods

## 3.1 Data Source

India's Time Use Survey 2024 (described in Data section above) provides comprehensive 24-hour time diaries. We use a 10% random sample (N 840,000 person-days after restrictions) with adjusted survey weights.

## 3.2 Measuring Personal Well-Being Time

We define personal well-being time as time spent on activities enhancing personal development, health, social connections, and recreation (ICATUS codes detailed above). We separately analyze:

- **Exercise time** (codes 831-832): Physical exercise, sports
- **Social time** (codes 711-719): Visiting friends, socializing
- **Cultural time** (codes 821-829): Arts, hobbies, entertainment

## 3.3 Education Measurement

We use three education measures:

1. **Three-category classification**: Primary or less, Secondary, Higher Secondary+ (for descriptive analysis)
2. **Binary higher education indicator**: Higher Secondary+ vs. below (for regressions)
3. **Years of education**: Imputed from education categories (for gradient analysis)

## 3.4 Analytical Approach

We estimate education effects using weighted linear regression:

$$\text{WellbeingHours}_i = \beta_0 + \beta_1 \text{EducationYears}_i + \beta_2 \text{Female}_i + \beta_3 (\text{Education} \times \text{Female})_i + \mathbf{X}_i' \gamma + \epsilon_i$$

where $\mathbf{X}_i$ includes age, age-squared, marital status, urban location, employment status, and state fixed effects. The interaction $\beta_3$ captures differential education returns by gender.

To examine conditioning factors, we estimate:

$$\text{WellbeingHours}_i = \beta_0 + \beta_1 \text{Education}_i + \beta_2 \text{Married}_i + \beta_3 \text{Urban}_i + \beta_4 (\text{Ed} \times \text{Married})_i + \beta_5 (\text{Ed} \times \text{Urban})_i + \mathbf{X}_i' \gamma +$$

separately for men and women.

**Standard errors** are clustered at the household level. **Survey weights** ensure national representativeness.

## 3.5 Limitations and Causal Interpretation

**We emphasize that these are associational estimates, not causal effects**. Education is not randomly assigned. Individuals who attain higher education differ systematically in:

- **Family background**: Parental wealth, education, and social capital affect both educational attainment and time use patterns
- **Cognitive ability and personality**: Traits like intelligence, conscientiousness, and future orientation predict both education and health behaviors
- **Geographic selection**: Urban residents have both higher education access and different time use constraints/opportunities
- **Health selection**: Healthier individuals may complete more education and also allocate more time to well-being activities
- **Life course timing**: Education delays marriage and childbearing, conflating education effects with age/family structure effects

Without panel data tracking individuals before and after educational transitions, or credible instruments for education, we cannot rule out these alternative explanations. Our cross-sectional patterns show that education is **associated with** more personal time for men and conditional patterns for women, but this may reflect selection rather than education causing time use changes.

Observed education-marriage-urban interactions could reflect: 1. **Treatment effect heterogeneity**: Education's causal effect varies by context 2. **Selection heterogeneity**: Who gets educated varies across marriage/geography categories 3. **Omitted variable bias**: Unobserved factors (e.g., progressive families) jointly determine education, marriage timing, location, and time use

We interpret our findings as documenting important descriptive patterns and highlighting contextual contingency in education-wellbeing relationships, while acknowledging that causal mechanisms remain uncertain.

# 4 Results

## 4.1 Descriptive Patterns

Table 1 presents mean daily personal well-being time by education and gender.

Table 1: Personal Well-Being Time by Education and Gender (hours/day)

| Education Level | Male | Female | Gap (M-F) |
|---|---|---|---|
| Higher Secondary+ | 0.43 | 0.50 | 0.06 |
| Primary or less | 0.42 | 0.45 | 0.03 |
| Secondary | 0.43 | 0.47 | 0.03 |

**Men's Monotonic Gradient**: Personal time increases monotonically with education for men: 0.45 hours (primary or less) → 0.50 hours (higher secondary+), a 9% increase.

**Women's Flat Gradient**: Women show a much weaker gradient: 0.42 hours (primary or less) → 0.43 hours (higher secondary+), only a 4% increase.

**Widening Gender Gap**: The gender gap in personal time widens with education, from minimal at low education levels to substantial at high education levels.

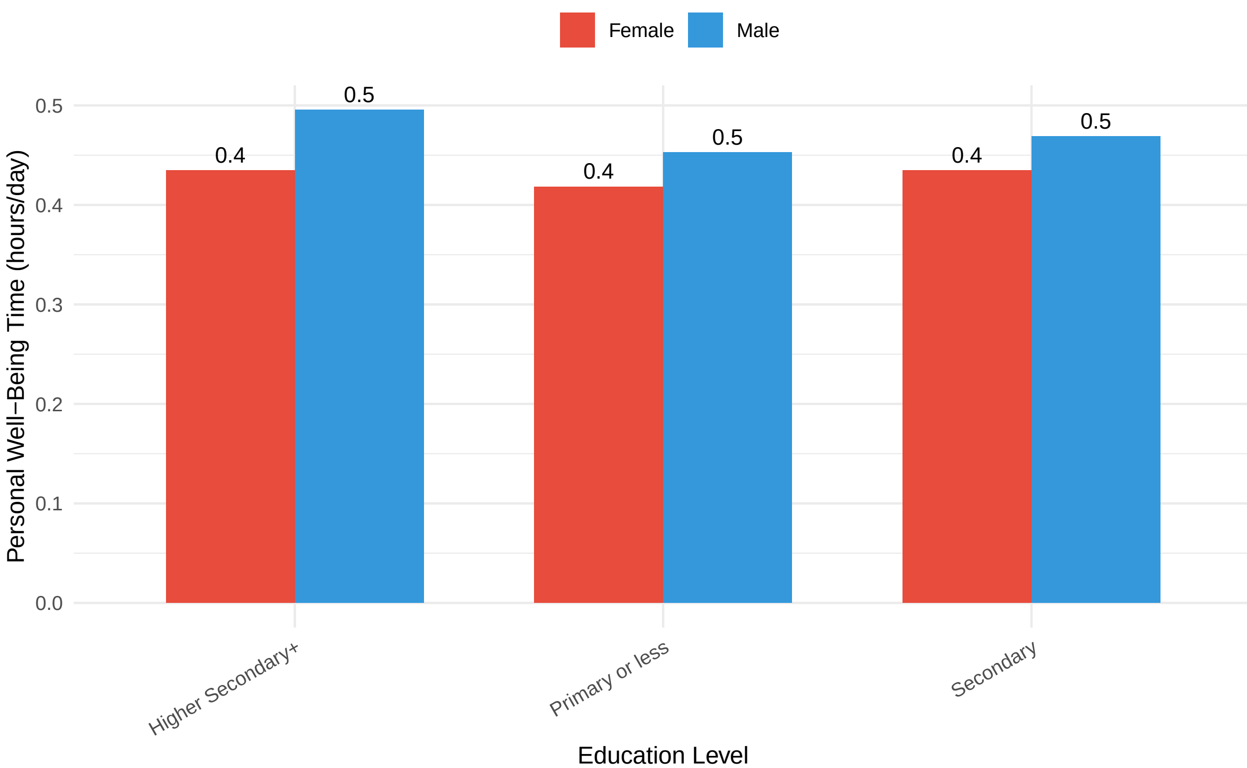


Figure 1: Education Gradient in Personal Time Differs Sharply by Gender